

Bike Index

Admin App — Designer Brief

Prepared for Kelsey · June 2026 · v1.0

Purpose & Background

Bike Index is used by university transportation teams to manage bike registration, impounding, and campus operations. The current admin experience is browser-only and requires too many steps to complete common field tasks. This brief outlines a redesign focused on two related surfaces:

- A simplified mobile admin app with a 3-button interface for field staff
- A Bike Show (bike detail page) that serves as the action hub for both mobile and desktop
- Desktop dashboard parity — the same streamlined flows should work on the desktop admin panel

The goal is to reduce friction for daily field tasks (impounding, messaging, scanning stickers) while keeping the full management experience available for managers at their desks.

The 3-Button Mobile Interface

The mobile admin app should have exactly three entry points. All roads lead to the Bike Show page.



SEARCH

Find by serial, name, email, or QR



ADD

Register new bike or log unregistered impound



SCAN

Scan sticker → jump to Bike Show

NOTE

Jeff, Ian, and other field staff also want to use the mobile interface when out scanning bikes — even if they have manager-level permissions. The app should serve both field use and quick lookups for all roles.

Search

Find bikes by serial number, owner name, owner email, or QR code scan. Returns a list; tapping a result opens the Bike Show page.

Add a Bike

Two modes: (1) Register a new bike normally, (2) Log an unregistered bike for impound with a photo. Photo upload must not block the user from immediately adding another bike — this is currently a blocking issue during high-volume impound sweeps.

Scan

Scan a Bike Index sticker. Immediately navigates to the Bike Show page for that bike. No intermediate steps.

Bike Show Page — The Action Hub

The Bike Show page is where staff act on a bike. It needs to surface all relevant actions without requiring navigation away. This page should work identically on mobile and desktop.

Actions that should live on Bike Show:

- Send a message (email and/or SMS) to the owner
- Impound the bike (see Impound section below for UX detail)
- Create or review parking notifications linked to this bike
- View photos attached to the bike or its notifications

NOTE

This is also a good opportunity to improve how Bike Show works for non-organization users (e.g. police, public). Consider what information and actions should be visible at each permission level.

Impound — UX Detail

Impound needs to be split out from the parking notification flow. It should be a distinct, fast action — not nested inside the notification action.

Parking Notification Types

The current notification system needs clearer type definitions. Current types:

- Appears abandoned
- Parked incorrectly
- Impound
- Other (consider renaming current "General message" to "Other")

Location Handling

Notifications and impounds currently offer "Use current location" or "Enter address manually". This should be updated to:

- Use existing location — if this bike already has a location from a prior notification
- Set new location — opens a map with a draggable pin; pin defaults to the user's current location

Impound Action

The impound button should mirror the same location UX as notifications (existing location or set new via map). It should be a single confirmation.

Role-Based Permissions

Three roles are in scope. The table below defines what each role can do.

Capability	Manager / Admin	Employee	Limited (RA etc.)
Manage org settings & users	✓	—	—
Bulk import / export	✓	✓	—
Impound a bike	✓	✓	—
Release impounded bikes	✓	✓	—
View all parking notifications sent	✓	✓	—
See graduation / sticker / recovery lists	✓	✓	—
E-vehicle actions	✓	✓	—
Search & message bike owners	✓	✓	✓
View student email	✓	✓	✓
Approve notifications before send	✓	—	—

SSO

Field staff want SSO so they don't have to remember another password. Magic sign-in should persist — staff should not be logged out between sessions.

User Stories by Area

The following stories are from the 2026 University Bike Index backlog. Stories marked Needs Significant Updates have existing functionality that requires redesign. Stories marked Not Implemented are net-new.

Impounding & Abandoned Bikes

Role	User Story	Status	Freq
Transportation Employee	I want to create an unregistered bike from a mobile flow with a photo so that there is a record of it in the admin backend	Not Implemented	Daily
Transportation Employee	I want the photo upload to not block adding additional bikes	Needs Updates	Daily
Transportation Employee	I want to impound an existing bike very quickly so that I can impound a bunch of bikes in a row	Needs Updates	Monthly
Campus Safety	I want to perform a spot check on students riding dangerously so that we have them in the system	Needs Updates	Daily
Transportation Manager	I want to easily manage impounded bikes coming in and out of the system so I can get them in to and out of my warehouse	Needs Updates	Monthly

Role	User Story	Status	Freq
Transportation Employee	I want to be able to see who and what role impounded or sent notifications	Needs Updates	Monthly
RA	I want to easily request impounding of a bike parked in an unsafe/illegal/annoying way so that it stops being like that	Not Implemented	Daily
Transportation Employee	I want to take a photo of a bike that looks abandoned so I can follow up after a defined period to impound it	Needs Updates	Daily
Transportation Employee	I want to review the photos and messages I sent so I can follow up on e-vehicles and impound them if still present	Not Implemented	Daily
Transportation Manager	I want to export impound information so that it is present in other systems we use	Needs Updates	Monthly

Sign Up & Onboarding (Move-In Day)

Role	User Story	Status	Freq
Transportation Employee	I want to register the bulk of vehicles on move-in day so that I have access to the highest number of vehicles	Needs Updates	Annually
Transportation Manager	I want to run a registration drive at the beginning of the year so that incoming students get into the system	Needs Updates	Annually
Transportation Manager	I want to easily distribute stickers in the most efficient manner possible so students can get them during registration	Needs Updates	Annually
Transportation Manager	I want a shareable training resource for Bike Index so my staff know their roles and abilities	Not Implemented	Annually
Transportation Manager	I want physical resources and assets so that I can create multiple registration touch points	Not Implemented	Monthly

Notifications & Messaging

Role	User Story	Status	Freq
Transportation Employee	I want to mass message segments of the population so that policy can be targeted and communicated	Needs Updates	Monthly
Transportation Employee	I want to send messages to students via email and phone so that they can deal with a problem	Needs Updates	Daily
Transportation Employee	I want to send messages notifying students about rack/parking closures so that they move their vehicles prior	Needs Updates	Monthly
Transportation Manager	I want to easily message users in bulk to share any university policy updates quickly with all bicycle users on campus	Needs Updates	Monthly
Transportation Employee	I want an admin app so that I don't have to login on browser repeatedly — keep me signed in!	Not Implemented	Daily

Role	User Story	Status	Freq
Transportation Employee	I want to add a photo to a notification without leaving the app	Not Implemented	Daily
Transportation Manager	I want to approve notifications created by lower-privilege admins prior to send	Not Implemented	Daily
Transportation Employee	I want to use SSO so that I don't have to remember another password	Not Implemented	Daily
Student	I want to be able to contact a known locale for information about biking on campus so I can follow the rules or reclaim my bike	Needs Updates	Monthly
Student	I want to easily see campus resources — bike index (e.g. report stolen) and campus (e.g. bike map)	Needs Updates	Monthly

E-Vehicle

Role	User Story	Status	Freq
RA	I want to check whether an e-vehicle is allowed to be in the dorm so that things don't light on fire	Not Implemented	Daily
Transportation Manager	I want to make students complete a safety training so that they ride safer on campus	Not Implemented	Monthly
Transportation Manager	I want to make e-vehicle policy known at point of sign-up so that there is a record of agreement	Not Implemented	Monthly
Transportation Employee	I want to see whether a student has completed safety training and agreed to policy	Not Implemented	Daily
Transportation Employee	I want to take a photo of an e-vehicle charging somewhere not allowed so I have a record of what I did	Needs Updates	Daily
Transportation Manager	I want to easily understand what e-vehicles are on campus so that I can quantify their existence	Needs Updates	Monthly
Transportation Manager	I want to impose a battery size limit on campus so there is a speed limit for e-vehicles	Not Implemented	Monthly
Transportation Manager	I want to impose an e-vehicle class restriction on campus so that commuting is safer	Not Implemented	Monthly
Student	I want to register my e-vehicle so that the school lets me charge it	Needs Updates	Annually
Student	I want to do the safety course for my e-vehicle so that the university lets me use it on campus	Not Implemented	Annually

Reporting & Offboarding

Role	User Story	Status	Freq
Transportation Manager	I want to understand the movement of bikes in and out of the system so I have a comprehensive view and can better manage all activities	Needs Updates	Annually

Role	User Story	Status	Freq
Sustainability Manager	I want a picture of all vehicles on campus and where they're located so I can understand modal shift and justify more bike infrastructure	Not Implemented	Annually
Transportation Manager	I want to run reports so that I can see bicycles' activity on campus	Not Implemented	Monthly
Transportation Manager	I want to pull sustainability metrics so that we can lobby for funding	Not Implemented	Annually
Student	I want to sell my bike easily at the end of the year	Not Implemented	Annually

Screenshots & Reference

Gavin will add annotated screenshots of the current admin interface here, for reference. Screenshots are provided for context only — the new design does not need to match the current layout.

**AREAS
TO
CAPTURE**

Registrations list · Bike Show page · Parking notifications flow · Impound list · Add a bike form · Mobile app current state